RECOGNITION PIPELINE

THREE-TIER ARCHITECTURE / METAMEDIUM

INPUT

raw stroke data

points[]
{x, y} × n
mousedown → mousemove → mouseup

FINGERPRINT EXTRACTION

getFingerprint() + enrichment

GEOMETRIC

- straightness — 0–1 direct/path
- aspectRatio — width / height
- isClosed — start↔end distance
- cornerCount — inflection points
- cornerAngles[] — degrees at corners
- pointCount — sample density

CURVATURE

- avgCurvature — Menger curvature mean
- maxCurvature — peak bending
- curvatureVar — organic vs uniform
- directionChain — Freeman 8-code
- chainLength — encoded path length

TOPOLOGICAL

- symmetry.V — vertical mirror 0–1
- symmetry.H — horizontal mirror 0–1
- selfIntersect — crossing count
- angleAnalysis — shape likeness
  - .rectLikeness — rectangle score
  - .triLikeness — triangle score

DIRECTION CHAIN ENCODING

example:
R R DR D D DL L L UL U U UR
→ closed loop (circle-like)

THREE-TIER RECOGNITION

TIER 1 INSTANT HEURISTICS

analyzeStrokeDetailed()
< 5ms | synchronous | always fires

CIRCLE	straightness < 0.4 && closed && aspect ≈ 1.0
LINE	straightness > 0.75 && !closed
RECTANGLE	0.5 < straight < 0.75 && closed && 4 corners
TRIANGLE	closed && 3 corners && triLikeness > 0.6
ARROW	!closed && 2–3 corners && directional
ARC	straightness < 0.6 && !closed && curved

confidence: 0% — 100%
80% auto-accept

parallel →

TIER 2 LIBRARY MATCH

findNearestLibraryMatch()
< 10ms | synchronous | if library exists

- user-defined shapes
- saved compositions
- custom vocabularies

fingerprint comparison:
euclidean distance
weighted features
threshold: 0.3

conditional
conf < 80%

TIER 3 LLM SEMANTIC

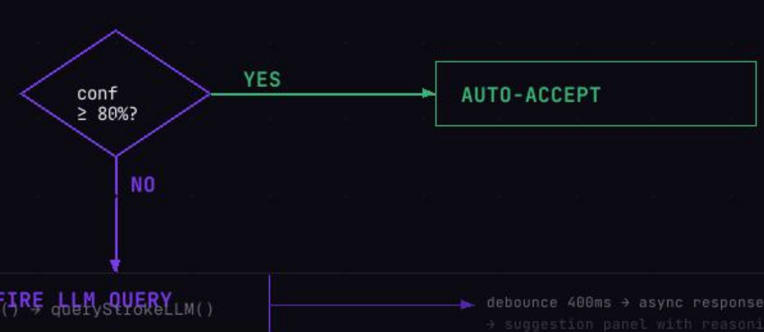
queryStrokeLLM()
~2s | async | confidence < 80%

- 400ms debounce
- enriched fingerprint
- spatial context (graph)
- scene relationships

returns:
label + confidence
reasoning chain
alternatives[]

CONFIDENCE GATE

when to invoke the LLM



skip LLM, save compute

debounce 400ms → async response
→ suggestion panel with reasoning

DATA FLOW TO LLM

serializeStroke() → queryStrokeLLM()

```
{
  fingerprint: {
    straightness: 0.23,
    aspectRatio: 0.95,
    isClosed: true,
    cornerCount: 0,
    cornerAngles: [],
    pointCount: 87,
    angleAnalysis: { rectLike: 0.1, ... }
  },
  curvature: {
    avg: 0.034, max: 0.051, var: 0.0008
  },
  symmetry: { V: 0.92, H: 0.89 },
  selfIntersections: 0,
  directionChain: "R DR D DL L UL U UR",
  canvasContext: [ ... ],
  spatialContext: [ ... ]
}
```

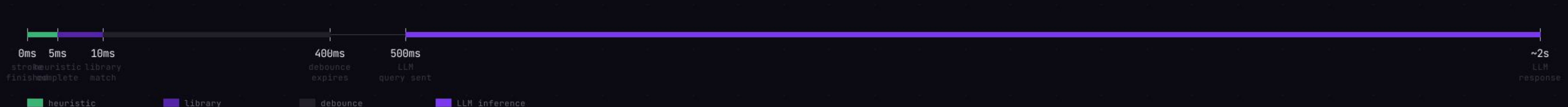
existing

NEW enriched

CONTEXT PAYLOAD

stroke data	fingerprint + curvature + topology
scene context	what else is on canvas
spatial graph	relationships between strokes
system prompt	interpretation guidelines
total tokens:	~800–1200 per query

TIMING



STROKE LIFECYCLE

FROM TOUCH TO UNDERSTANDING / METAMEDIUM

1 CAPTURE mousedown → 2 COLLECT mousemove → 3 FINALIZE mouseup → 4 EXTRACT fingerprint → 5 ANALYZE heuristics → 6 SUGGEST panel update → 7 LLM conditional → 8 RESOLVE accept/reject

PHASES 1–2

input capture

EVENT HANDLERS

canvas.mousedown	state.isDrawing = true	begin stroke
state.currentStroke = []		
push first point		
canvas.mousemove	if (!state.isDrawing) return	collect points
push {x, y} to currentStroke		
render live stroke		
state.isDrawing	persists across events	drawing flag
state.currentStroke	point array, reset on finalize	raw data

PHASE 3

stroke finalization — the mouseup fix

CANVAS MOUSEUP

mouse released ON canvas

1. set _canvasMouseUpHandled = true
2. push currentStroke → strokes[]
3. analyzeStrokeDetailed()
4. updateSuggestionPanel()
5. conditional LLM query
6. reset isDrawing + currentStroke

DOCUMENT MOUSEUP

mouse released OUTSIDE canvas

1. check _canvasMouseUpHandled flag
   → if true: reset flag, return
   → if false: proceed (outside release)
2. check isDrawing
3. same finalization as canvas handler
4. stroke completes properly

_canvasMouseUpHandled

prevents double-processing

PROBLEM:
mouseup on canvas fires
BOTH canvas + document
handlers (event bubbling)

SOLUTION:
canvas handler sets flag
document handler checks it
→ skips if already handled

mouseleave: no-op — document mouseup handles finalization wherever release occurs

PHASES 4–5

extraction & analysis (synchronous, < 10ms)

raw stroke points[] → getFingerprint() 11 features → enrichment +curvature +sym → analyzeDetailed() all shapes → results[] {label, score}

GEOMETRIC	ANGULAR	CURVATURE	TOPOLOGY
straightness 0–1	cornerAngles[] deg	avgCurvature float	symmetry.V 0–1
aspectRatio	rectLikeness 0–1	maxCurvature float	symmetry.H 0–1
isClosed bool	triLikeness 0–1	curvatureVar float	selfIntersect int
cornerCount int	consistency 0–1	directionChain str	pointCount int

PHASE 6

suggestion panel display

SUGGESTION PANEL

TOP MATCH
Circle 85% [Accept]
← heuristic match (instant)

● AI analyzing...
→ appears here with reasoning
→ chip alternatives inline
← LLM suggestion (async) moved UP for comparison

OTHER CANDIDATES
- Ellipse 72%
- Oval 65%
- Arc 41%

← lower-ranked matches

REASONING
straightness=0.23 (curved)
closed=true, aspect=0.95
symmetry V=0.92 H=0.89

DISAGREEMENT DETECTED

Heuristic: "Arc"
LLM: "Wave"
red accent border signals
AI sees something different
= high-value interpretation

PHASE 7

conditional LLM query

topConfidence = results[0].score → < 80 ?

- NO (≥80%) → AUTO-ACCEPT → skip LLM entirely, save GPU compute
- YES → queryStrokeLLM() → 400ms debounce → fetch() → parse JSON → appendLLMSuggestion()

PHASE 8

resolution — user decides

ACCEPT

context[i] = label

stroke turns blue (#0066ff)
label stored in context[]
library updated

REJECT

shape dismissed

stroke remains gray
no label applied
try different suggestion

RENAME

custom vocabulary

user types custom name
stored as new vocabulary
teaches the system

STATE TRANSITIONS

IDLE —mousedown→ DRAWING —mouseup→ FINALIZING —sync→ ANALYZING —display→ SUGGESTING —user action→ RESOLVING

RESOLVING → IDLE: reset → ready for next stroke

SPATIAL GRAPH

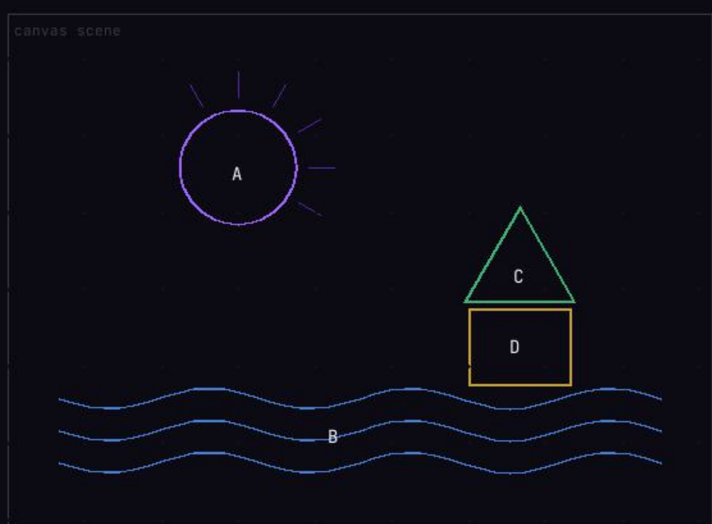
COMPOSITIONAL CONTEXT / METAMEDIUM

"the graph relationships are the genius of our little demo"
"they give the model a grounded context so it's not just looking at compressed images"

serializeScene()

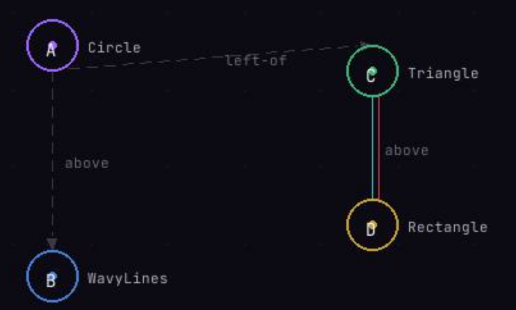
building the spatial context graph

canvas scene



RELATIONSHIP GRAPH

edges computed by serializeScene()



EDGE TYPES

- - -	proximity	bounding boxes within threshold
—	touching	edges within 5px
—	overlap	bounding boxes intersect
███	containment	one fully inside another

SERIALIZED CONTEXT

what the LLM receives


```
canvasContext:
[
  { index: 0, label: "Circle", confidence: 85,
    bounds: {x:135, y:75, w:90, h:90},
    fingerprint: { straight:0.23, closed:true, ... } },
  { index: 1, label: "WavyLines", confidence: 62,
    bounds: {x:40, y:280, w:470, h:70}, ... },
  { index: 2, label: "Triangle", confidence: 78, ... },
  { index: 3, label: "Rectangle", confidence: 91, ... }
]
```

```
spatialContext:
[
  { from: 0, to: 1,
    relationship: "above",
    distance: 160,
    direction: "below" },

  { from: 2, to: 3,
    relationship: "overlapping",
    overlap: 0.4 },

  { from: 0, to: 2,
    relationship: "near",
    direction: "right" }
]
```


COMPOSITIONAL REASONING

how spatial context enables semantic interpretation

SCENE: SUN OVER WATER

circle.above(wavyLines)
→ sun over water
→ landscape / seascape

SCENE: FACE

smallCircles.inside(largeCircle)
→ eyes inside head
→ face / character

SCENE: HOUSE

triangle.above(rectangle)
→ roof on walls
→ house / building

SCENE: RADIAL PATTERN

lines.radiatingFrom(circle)
→ sun / flower / star
→ depends on line style

WHY SPATIAL CONTEXT MATTERS

WITHOUT SPATIAL CONTEXT

LLM receives isolated fingerprints

"circle" → circle, ring, dot, wheel, ball...
"wavy line" → wave, snake, river, squiggle...
"triangle" → triangle, arrow, mountain, roof...
"rectangle" → box, door, window, wall, screen...

NO compositional understanding
EVERY shape is ambiguous in isolation

VS

WITH SPATIAL CONTEXT

LLM receives fingerprints + relationships

circle ABOVE wavyLines → "sun" (not ball)
smallCircles INSIDE bigCircle → "eyes" (face)
triangle TOUCHING rectangle → "roof" (house)
lines RADIATING FROM circle → "sun rays"

GROUNDED semantic reasoning
Context disambiguates meaning

PROMPT ARCHITECTURE

what the system prompt teaches

ROLE
You are a drawing recognition assista
Your value: SEMANTIC interpretation.

FEATURES
Interpret fingerprint values:
straightness, curvature, symmetry, ...

ORGANIC SHAPES
High curvature + closed + symmetric → circle
Low straightness + open + wavy → wave

COMPOSITION
Use canvasContext + spatialContext:
circle above waves = sun over water

OUTPUT
Return JSON: label, confidence,
reasoning (cite features), alternatives[]

COMPLETE DATA FLOW

DRAW	~500 pts →	FINGERPRINT	17 floats →	SERIALIZE	~800 tokens →	SPATIAL GRAPH	+edges →	LLM	JSON out →	SUGGEST
user input		17 features		stroke + scene		relationships		semantic inference		grounded result

COMPRESSION → CONTEXT → MEANING

500 raw points become 17 features + spatial edges = grounded understanding